

ASSIGNMENT 7 REPORT

FORMFLOW – MULTI-SCREEN FORM APPLICATION IN FLUTTER

[Github Link](#)

1. INTRODUCTION

Assignment 7 focuses on developing a multi-screen Flutter application using forms, form validation, and named-route navigation. For this assignment, I developed an application called “FormFlow” with the tagline “Simple Forms. Better Futures.”

The application demonstrates a complete registration flow starting from a Home Screen, continuing to a Registration Form, and ending with a Detail/Success Screen. The project also includes responsive UI, animations, reusable widgets, dynamic data, and a consistent application theme.

The main purpose of this assignment was to understand how Flutter handles user input, validates form data, navigates between multiple screens, and transfers information from one screen to another.

2. OBJECTIVES

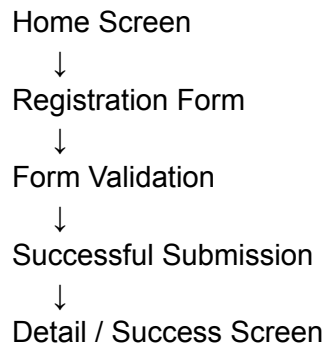
The main objectives of this assignment were:

- Create a multi-screen Flutter application.
- Design a Home Screen.
- Create a Registration Form Screen.
- Implement form validation.
- Validate required fields.
- Validate email format.
- Validate password requirements.
- Confirm password matching.
- Use named routes for navigation.
- Pass data between screens.
- Display submitted information dynamically.
- Create a responsive interface.
- Add animations and modern UI/UX.
- Organize the application into separate files and screens.

3. APPLICATION OVERVIEW

FormFlow is a registration-based Flutter application that demonstrates how multiple screens can work together as one complete user flow.

The application follows this flow:



The Home Screen introduces the application and provides a “Get Started” button. When the user selects it, the Registration Screen opens.

The Registration Screen collects the user’s personal information and validates the entered data. If all fields are valid, the application proceeds to the Detail Screen.

The Detail Screen displays the submitted information and confirms that registration was successful.

5. HOME SCREEN

The Home Screen acts as the introduction to FormFlow. Instead of using a simple academic-style interface, the screen was designed with a modern and premium visual style.

The Home Screen contains:

- FormFlow logo and tagline.
- Flutter Assignment 07 badge.
- Main hero heading.
- Application description.
- Get Started button.
- Learn More button.
- Animated phone mockup.
- Floating icons.
- Decorative background elements.
- Feature cards.
- Registration call-to-action section.
- Quote section.

The main heading is:

“Build Your Future With FormFlow”

The Home Screen is responsive. On desktop screens, the hero section uses a two-column layout with the text on one side and the phone illustration on the other. On smaller screens, the content is arranged vertically.

The Learn More button also opens a bottom sheet containing information about the application.

6. REGISTRATION FORM

The Registration Screen is the main functional part of the application.

The form contains four fields:

1. Full Name
2. Email Address
3. Password
4. Confirm Password

Flutter's Form and TextFormField widgets are used to collect and validate the input.

TextEditingController is used to access the values entered by the user.

The form uses a GlobalKey<FormState> to control and validate the complete form before submission.

7. FORM VALIDATION

Validation is one of the main requirements of the assignment.

Each field has its own validation rule.

Full Name:

The name field is required and must contain at least two characters.

Email:

The email field is required and is checked using an email validation pattern.

Password:

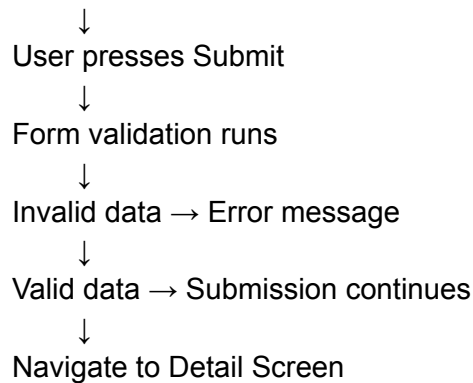
The password field is required and must contain at least six characters.

Confirm Password:

The confirmation field is required and must match the password entered by the user.

The validation flow is:

User enters information



For example, if the user leaves the name empty, a validation message is displayed. If an invalid email is entered, the form does not submit. Similarly, a password shorter than six characters or a mismatched confirmation password is rejected.

This prevents invalid information from being passed to the next screen.

8. NAMED ROUTE NAVIGATION

Named routes are used to navigate between the three main screens.

The routes used are:

/ → Home Screen
/form → Registration Screen
/details → Detail Screen

The application uses `onGenerateRoute` in `main.dart` to create the routes.

Custom page transitions were also added:

- Fade transition for the Home Screen.
- Slide-up and fade transition for the Registration Screen.
- Scale and fade transition for the Detail Screen.

These transitions provide a smoother and more modern navigation experience.

After successful form validation, the Registration Screen uses `Navigator.pushNamed` to open the Detail Screen.

9. PASSING DATA BETWEEN SCREENS

The application demonstrates how data can be passed from one screen to another.

After successful registration, the user's name and email are passed as route arguments.

The Detail Screen receives these arguments and displays them dynamically.

The application does not pass or display the password on the Detail Screen. Only the information required for the registration confirmation is transferred.

The registration date is also generated dynamically using the current date.

10. DETAIL / SUCCESS SCREEN

The Detail Screen represents the successful completion of the registration process.

It contains:

- Animated success icon.
- Welcome Aboard message.
- Registration confirmation.
- Full Name.
- Email Address.
- Registration Date.
- Account Status.
- Success message.
- Go to Home button.
- Register Again button.
- Motivational quote.

The user's information displayed on this screen comes from the registration form rather than being hardcoded.

The registration status is displayed as "Registered".

The screen also includes entrance animations, floating effects, and a confetti effect to make the successful registration experience more engaging.

11. RESPONSIVE DESIGN

Responsiveness was an important part of the project.

Flutter widgets such as LayoutBuilder, MediaQuery, Expanded, Flexible, and Wrap were used to create layouts that adapt to different screen sizes.

On desktop screens:

- The hero section uses two columns.
- Feature cards appear in a row.
- More spacing is provided.
- Larger typography is used.

On mobile screens:

- The hero content becomes vertically stacked.
- Feature cards are arranged vertically.
- Buttons can wrap when required.
- Smaller spacing and phone dimensions are used.

The application was tested at desktop, tablet, and mobile screen sizes to ensure that the interface remained usable and did not produce unwanted horizontal overflow.

12. UI/UX DESIGN

The application was designed to look modern and different from a basic form assignment.

The visual design uses:

- Purple, blue, and pink gradients.
- Rounded cards.
- Soft backgrounds.
- Subtle shadows.
- Modern typography.
- Animated elements.
- Clear visual hierarchy.
- Consistent spacing.
- Responsive layouts.
- Interactive buttons.

A common AppTheme class was created so that colors, typography, borders, and other design elements remain consistent across the application.

Animations were used for page transitions, background elements, hero content, floating objects, buttons, and the success screen.

13. DYNAMIC FEATURES

Several parts of FormFlow are dynamic rather than static.

The application dynamically:

- Reads user input.
- Validates form fields.
- Displays validation errors.
- Shows a submission/loading state.
- Passes registration data between screens.
- Displays the user's name and email.

- Generates the current registration date.
- Changes its layout according to screen size.
- Animates different UI components.

14. TESTING

The application was tested using different user input conditions and screen sizes.

Home Screen testing:

- Application opens successfully.
- Get Started opens the Registration Screen.
- Learn More opens the information bottom sheet.
- Responsive layout works at different widths.

Registration testing:

- Empty fields are rejected.
- Invalid email addresses are rejected.
- Short passwords are rejected.
- Mismatched passwords are rejected.
- Valid information allows submission.
- Loading state appears during submission.

Detail Screen testing:

- Name is displayed correctly.
- Email is displayed correctly.
- Current registration date is displayed.
- Registered status is displayed.
- Go to Home returns to the Home Screen.
- Register Again opens the Registration Screen.

Responsive testing was also performed on desktop, tablet, and mobile layouts.

15. KEY CONCEPTS LEARNED

This assignment helped me understand several important Flutter concepts.

Forms:

I learned how Form and TextFormField are used to collect user information.

Validation:

I learned how validator functions can check required fields, email formats, password length, and password confirmation.

Controllers:

I learned how `TextEditingController` can be used to read values entered into text fields.

Named Routes:

I learned how named routes can organize navigation between different screens.

Route Arguments:

I learned how data can be passed from one screen to another using `Navigator.pushNamed` and route arguments.

StatefulWidget:

I learned how `StatefulWidget` is useful when the UI needs to change because of user interaction or animations.

Animations:

I learned how `AnimationController`, `TweenAnimationBuilder`, `AnimatedBuilder`, `FadeTransition`, `SlideTransition`, and `ScaleTransition` can be used to create smooth UI effects.

Responsive Design:

I learned how `LayoutBuilder` and `MediaQuery` can be used to create interfaces that adapt to different screen sizes.

Theming:

I learned how a shared theme can maintain consistent colors, typography, and component styling throughout an application.

16. RESULT

The final `FormFlow` application successfully implements the major requirements of Assignment 7.

The application includes:

- Home Screen.
- Registration Form Screen.
- Detail / Success Screen.
- Named-route navigation.
- Required-field validation.
- Email validation.
- Password validation.
- Confirm password validation.
- Dynamic data transfer.
- Responsive layouts.
- Animations.
- Modern UI/UX.
- Shared application theme.

The complete registration flow works successfully from the Home Screen to the Registration Screen and finally to the Detail Screen.